

Solutions – Quiz 1

ECON 101: Macroeconomics

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Instructions

Answer all questions. Show all work for numerical problems. Multiple-choice questions have one correct answer.

Part 1: Multiple Choice (2 points each)

1. Economics is best defined as the study of:
 - A. How governments set tax rates
 - B. How to eliminate scarcity
 - C. The choices people make to attain their goals, given scarce resources
 - D. How firms maximize profits only
2. Which of the following is an example of a **positive economic statement**?
 - A. The Bank of Canada should lower interest rates.
 - B. A higher minimum wage will lead to higher unemployment among teenagers.
 - C. Canada ought to spend more on public health care.
 - D. Equity should be prioritized over efficiency.
3. In the economic problem of scarcity:
 - A. Wants are limited, resources are unlimited.
 - B. Both wants and resources are unlimited.
 - C. Wants are unlimited, resources are limited.
 - D. Both wants and resources are limited.

4. The **Production Possibilities Frontier (PPF)** is bowed outward because:
 - A. Opportunity costs are constant.
 - B. Resources are not equally productive in all uses.
 - C. Scarcity does not apply.
 - D. Technology is fixed.
5. If Canada can produce wheat more efficiently than Japan, while Japan can produce electronics more efficiently than Canada, trade between the two countries can be mutually beneficial due to:
 - A. Absolute advantage
 - B. Comparative advantage
 - C. Specialization without opportunity costs
 - D. Increasing marginal costs

Part 2: Numerical Questions

6. Suppose a student has 10 hours per week to allocate between working (earning \$15/hour) and studying. What is the **opportunity cost** of one additional hour of studying? **(2 points)**
7. Consider the following simplified PPF data:

Choice	Cars	Computers
A	0	100
B	20	80
C	40	60
D	60	40
E	80	20
F	100	0

- (a) What is the opportunity cost of increasing car production from 20 to 40? **(2 points)**
 - (b) Does this country face increasing, decreasing, or constant opportunity costs? Explain. **(2 points)**
8. Country A can produce 50 tons of steel or 100 tons of corn with all its resources. Country B can produce 60 tons of steel or 120 tons of corn.
 - (a) Which country has the **absolute advantage** in each good? **(2 points)**
 - (b) Which country has the **comparative advantage** in steel? **(2 points)**

Answer Key

Part 1: Multiple Choice

1. C
2. B
3. C
4. B
5. B

Part 2: Numerical

6. Opportunity cost of 1 hour studying = \$15 lost wages.
- 7a. Going from 20→40 cars reduces computers from 80→60 → opp. cost = 20 computers.
- 7b. Opportunity cost is constant (linear trade-off).
- 8a. Absolute advantage: steel = B, corn = B.
- 8b. Comparative advantage: compute opp. cost. A: 1 steel = 2 corn. B: 1 steel = 2 corn.
Both same $\Rightarrow$ no comparative advantage (special case).